

2-Mark Questions & Answers -- DBMS Module 4: Relational Database Design

Q1. Define 1NF, 2NF, 3NF, and BCNF.

- * 1NF: A relation where all attribute values are atomic (no repeating groups/multivalued cells).

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Q2. What is partial functional dependency

A Partial Functional Dependency occurs when a non-prime attribute is functionally dependent on only a proper subset of a composite candidate key (e.g., in key AB, if $A \rightarrow C$, C is partially dependent).

Q3. Define transitive dependency with an example.

A Transitive Dependency occurs when a non-prime attribute determines another non-prime attribute through a primary key ($A \rightarrow B$ and $B \rightarrow C$, therefore $A \rightarrow C$).

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Q4. What is a multivalued dependency (MVD)

A Multivalued Dependency (MVD) $X \twoheadrightarrow Y$ exists when the presence of two tuples sharing attribute X implies that attribute Y is completely independent of the remaining attributes Z in the relation.

Q5. What is a lossless-join decomposition

A decomposition of relation R into R1 and R2 is lossless-join if $R = R1 \bowtie R2$ (joining R1 and R2 produces the exact original tuples without creating spurious/fake records).